

3	D	Using relative speed of approach = relative speed of separation $u_1 - u_2 = v_2 - v_1$ $0.10c - -0.040c = v_2 - v_1$ $v_2 = 0.14c + v_1 \text{ --- (1)}$

[Turn over

Qn	Ans	Solution
		Using conversation of linear momentum $1u \times 0.01c + 4u \times -0.040c = 1u \times v_1 + 4u \times v_2$ $0.01c - 0.16c = v_1 + 4v_2$ $-0.060c = v_1 + 4v_2 \text{ --- (2)}$ Substitute eqn (1) into (2) $-0.060c = v_1 + 4 \cdot 0.14c + v_1$ $v_1 = -0.124c$ $v_2 = +0.016c$
1	6	If V_{ice} is volume of ice cube